

Research paper

Decentralized peer-to-peer energy trading in microgrids: Leveraging blockchain technology and smart contracts

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ABSTRACT

Microgrids are regarded as vital components in contemporary realm of energy system improvement, resilience, and sustainability. In this paper a novel decentralized peer-to-peer energy trading system leveraging blockchain technology is proposed. The proposed model not only demonstrates the implementation of blockchain technology in microgrids but also transforms the energy sector by emphasizing decentralization, security and efficiency. This research aims to enhance the energy trading system by including smart contracts written on the Ethereum network. Energy Token and Demand Response contracts are integrated to enable dynamic interactions inside microgrids, which leads to a transparent, secure, and efficient energy trading system. Moreover, automation of energy transactions and elimination of intermediaries ensure cost effectiveness and utilization of excess energy potentially reduce dependency on the main grid. A microgrid system is designed in Simulink for distributed energy trading. Energy credits are represented by standard ERC-20 digital token and then demand response contract dynamically adjusts rewards and energy prices based on energy production and consumption. the ERC-20 contract manages the token transaction and then demand response contract enforces governance rules. The purpose of a demand response contract is to enable the automatic and efficient control of energy usage in response to changing demand and supply conditions. The use of the Web3 library further facilitates a direct and smooth connection between the blockchain network and microgrids. The results demonstrate the successful implementation of both smart contracts, making trading possible at noon when the combined generation from solar array and energy management system exceed the energy demand.

1. Introduction

Microgrids play a crucial role in the future power distribution system. Microgrids improve energy resilience by operating independently during grid failure or integrating with the main grid. In an era marked by rising frequency of climate relates issues, transition toward renewable energy resources is accelerating. For a future solution involving decentralizing power production, local controls are required in addition to utility providers. Decentralized microgrids produce electricity on a local level incorporating renewable energy sources. This minimizes dependency on centralized power plants and supports a more sustainable and environmentally friendly energy system. Renewable energy sources (RES) have emerged as a viable option for meeting rising energy demand, alleviating climate change, and contributing to long-term development. The integration of these ways is carried out in a distributed manner via microgrid systems; this provides a set of technological solutions that allow information exchange between the consumers and

the distributed generation centers (Mengelkamp et al., 2018). These systems are increasingly being integrated with renewable energy sources such as solar and wind power, highlighting their importance in the transition to greener energy solutions. However, managing these decentralized networks presents unique challenges, particularly in terms of energy distribution, transaction security, and real-time responsiveness to supply and demand fluctuations.

In recent years, improvements in blockchain technology have offered potential answers to these problems. Cryptocurrencies were originally based on blockchain technology, but it's applications now extend across various domains. Energy management is one area where the benefits are being perceived. Blockchain technology establishes a decentralized and transparent ledger for recording transactions over a computer network. The idea of digital tokens was brought to the commercial and economic world by blockchain-based distributed ledger technology, which made it possible to track digital assets and fairly allocate profits to the real creators and value-added service providers (YoungLee).

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Decentralization eliminates the need for a governing body in energy systems, which in turn reduces the number of possible failure points while simultaneously increasing the reliability of financial records. Again, consensus and transparency are provided by making a copy of the ledger available to every participant in the network. Distributed ledger technology enables power exchange between individual users in a decentralized energy grid. Without the need for middlemen, members of a microgrid can transact energy safely and directly with one another. Reduced transaction costs and increased customer agency over their power source are two benefits of this simplified energy trading procedure.

Peer-to-Peer (P2P) networks allow participants to act as both clients and servers, allowing them to share assets while operating as servers. This is a highly prevalent feature of blockchain technology and can be integrated in micro-grid for effective energy sharing. When a customer needs more energy, for instance, the P2P network enables them to get it from the closest source—which could be their neighbor who is using solar panels to generate energy. This is known as load-balancing (Pee et al., 2019).

An additional approach is to use smart contracts on a blockchain, which provide unique opportunities for automating and securing energy transactions. Smart contracts have the ability to fundamentally transform microgrid operations. Smart contracts, are programs that run autonomously on a blockchain, greatly facilitate the automation of energy transactions inside microgrids. When certain conditions are satisfied, the contracts' preset rules and logic trigger the automatic implementation of the agreements and reduces human intervention in the transaction procedure and minimizes costs of political risk or default. Moreover, because blockchain is distributed and unalterable it provides a secure repository for energy transactions and provide more reliable and efficient decentralized energy management system (Xue et al., 2017; Xu et al., 2022a).

The current energy management system in microgrids encounter several challenges including inefficient energy distribution, transparency in transactions and lack in flexibility in demand response. Consequently, these issues cause wastage of energy, mistrust amongst stakeholders, higher cost and system unreliability. Furthermore, integrating real-world microgrid data with blockchain technology is a significant barrier. To address aforementioned problems, in our proposed model, smart contracts are written on Ethereum network for transparent and fair transactions and to improve efficiency of microgrids. Moreover, the microgrid employs a tokenized energy system utilizing ERC20 tokens serve as digital entities that manage distribution of power based on consumption demands. Additionally, the integration of Web3 framework enables a direct and seamless interface between microgrid and blockchain network. The objective is to empower consumers to adopt their energy usage in real-time, optimizing the microgrid's efficiency and flexibility to meet the evolving demands of the energy market.

The rest of the paper is divided into six parts. Related work is mentioned in Section 2. Preliminary aspects of proposed model are mentioned in Section 3. System's architecture is presented in Section 4. Section 5 describes results and discussion while conclusion is drawn in Section 6.

2. Related work

Peer-to-peer (P2P) energy trading systems have displayed keen interest in blockchain as it provides a decentralized and encrypted environment. In (Niloy et al., 2021), a power trading system is presented that enables consumers to actively participate in the power grid by contributing renewable energy sources. By including a smart microgrid, the system aims to maintain its functionality even when disconnected from the national grid, thus achieving self-sufficiency. Contributors will generate income by distributing the generated energy using a smart contract. In (Crespo-Vazquez et al., 2020), concept for a community-based local energy market in which members can engage in

energy trading with each other through a local pool is proposed. In (Xu et al., 2022b) a privacy-preserving scheme based on blockchain technology is introduced, which ensures secure access by employing attribute-based encryption and enables the verification and transmission of blockchain data via homomorphic encryption¹. In (Muzumdar et al., 2021) authors proposed smart contracts that store data linked to energy trade on an Ethereum blockchain platform using a proof-of-stake (PoS) consensus method and they used Vickrey-Clarke-Grove (Vickrey auction) approach for incentive energy trading. The suggested system provides reliable and incentive-based energy trading.

The notion of digital tokens was brought to business and the economy by blockchain-based distributed ledger technology, which made it possible to track digital assets and fairly divide money to the real producers and service users who provide value. The concept of digital tokens is discussed by authors in (YoungLee, 2019). In (Todorean et al., 2021) decentralized energy market peer-to-peer trade as examples of physical asset digitalization using blockchain technology is discussed. A P2P energy trading system is modelled where prosumers' excess energy is digitally stored using the Lockable ERC20 token. Until the energy is delivered and settled using the data from the energy meters, the energy tokens that were committed in the blockchain transactions will remain locked. In (de Diego et al., 2021) authors have proposed a framework for energy trading using blockchain technology. The system is based on a sealed bid auction type, which ensures a secure, reliable, and private energy exchange mechanism. It is designed specifically for the stakeholders of an islanded part of the grid, who form an energy community. In addition, this system employs blockchain solutions to track the safety of smart grid devices, allowing it to stop transactions involving malicious individuals or compromised devices.

In (Mitrea et al., 2023), the implementation of secure multi-party computation (MPC) is employed in Ethereum to provide anonymity on a peer-to-peer energy flexibility market. This is achieved by combining the flexibility ordering data of the prosumers and securely storing it on the blockchain. A method of encoding orders in the energy market is provided to conceal the quantity of energy being exchanged. This is achieved by grouping prosumers together, dividing the energy from bids and offers, and generating orders at the group level. In (Xiong et al., 2024), an optimal scheduling framework for integrated energy systems is presented that takes into account multi-energy sharing and integrated demand response. The objective is to fully exploit the energy resources of these systems, achieve efficient utilization, and enhance flexibility on the demand side while also considering local energy consumption. A collaborative operational framework is built for numerous integrated energy systems in order to reduce the overall cost, utilizing the principles of Nash bargaining theory. An innovative framework TokenGreen, is mentioned in (Naik et al., 2024), which utilizes blockchain technology and nonfungible tokens (NFTs) to allow individuals to acquire ownership of energy assets by investing in distributed energy generation, and clean energy systems and distribution. This approach promotes trust and transparency among the participants. The proposed architecture utilizes the Ethereum virtual machine (EVM), ERC-721 non-fungible tokens (NFTs), interplanetary file system (IPFS), and solidity smart contracts to establish a marketplace for energy based on NFTs. Based on blockchain technology and smart contracts, a potential solution for energy trading is presented in (Menon and Aarthi, 2023). An algorithm is executed on the Ethereum platform, which is connected to Ganache. The application was developed using the Solidity programming language and was tested on the Ethereum Virtual Machine (EVM) platform. The main objective of this technique is to fulfill the need for energy, thereby creating a dynamic and decentralized energy market. The transaction will be conducted in a secure manner, resulting in reduced transaction costs and shorter settlement times.

Despite the numerous studies conducted on decentralized energy trading systems, existing systems face limitations like partial centralization, high transaction cost as well as challenges of security and scalability, which need to be addressed. Moreover, none of the previous

work provides a practical solution for direct connection between microgrid and blockchain network. This paper, a novel approach for decentralized energy trading through tokenization of energy credits is presented where excess energy is automatically traded according to rules defined in smart contracts. Our system is fully decentralized, automated and provide secure environment for energy trading. In this system, energy tokens are utilized for the purpose of trading, all transactions are recorded in blockchain. Our proposed solution, reduce cost, settlement time and eliminate intermediaries.

3. Preliminaries

3.1. Peer to peer blockchain based energy trading

In recent years the energy distribution systems have been transformed by the rise of distributed energy resources. Concurrently with the profound revolution in energy production and consumption, conventional energy consumers are transitioning into prosumers. Prosumers' electricity generation is unpredictable because to fluctuating sunlight and temperature. There are multiple solutions available to prosumers that possess an excess electrical energy supply. The energy can be stored for later consumption,

exported to the grid, or sold to other customers. The direct exchange of energy between consumers and prosumers is referred as peer-to-peer (P2P) energy trading (Gurmani et al., 2019). Fig. 1 illustrates peer-to-peer energy trading model, which demonstrations four main contributors: consumers, prosumers, the power company, and power sharing coordinator. The trading arrows and energy arrows reflect the energy and money exchanged between prosumers and consumers, respectively. A prosumer has the ability to vend electricity to either a consumer or another prosumer. The complete negotiation process is conducted through a platform that serves as power exchange coordinator. The unidirectional trade arrows indicate that consumers can solely obtain energy from the power sharing coordinator. The bidirectional trading arrows indicate that prosumers have the ability to both purchase and sell electricity to the energy sharing coordinator.

Blockchain-based peer-to-peer electricity trading markets have potential to supplant intermediaries with a decentralized network of validator nodes, commonly referred to as miners. Ledger's integrity is ensured and authenticated by miner's collaboration. The maintenance of the network, which involves verifying, storing, and updating replicas of the electricity transaction ledger, consumes the computational resources of miners (Zheng et al., 2017). Distributed Ledger Technology (DLT) is a decentralized peer-to-peer (P2P) network that maintains a distributed database (ledger) that is nearly immutable. To validate and order transactions, DLT uses consensus mechanisms, cryptography and back-referencing blocks and peer's identity is protected by providing a cryptographic pseudo identity to each peer, to start a transaction. All peers can perceive the transactions that users have started, and they are grouped together into a single block. Peer nodes use distributed consensus and cryptography to verify this block of transactions. Blocks are added to the chains kept at each mode after they are validated, making it nearly hard to change once they are. In the blockchain, a smart contract is a contract that binds the parties in accordance with the established policies. It is linked to its preset executable code, which runs when a message is received to its address, and it has its own private storage (Damisa et al., 2022). Fig. 2 shows the overview of blockchain based energy transaction.

3.2. Smart contracts

A smart contract is blockchain-based executable code designed to enforce, facilitate, and execute the provisions of an agreement. A smart contract's primary objective is to execute the terms of an agreement automatically upon the fulfillment of the specified conditions. Consequently, transaction charges are considerably low as compared to conventional systems which require a third party that can be trusted to carry out the terms of an agreement. A smart contract possesses three fundamental components: an account balance, private storage, and executable code. The state of the contract includes both the storage and the balance of the contract and it is kept on the blockchain and is modified whenever the contract is executed. Fig. 3 illustrates the smart

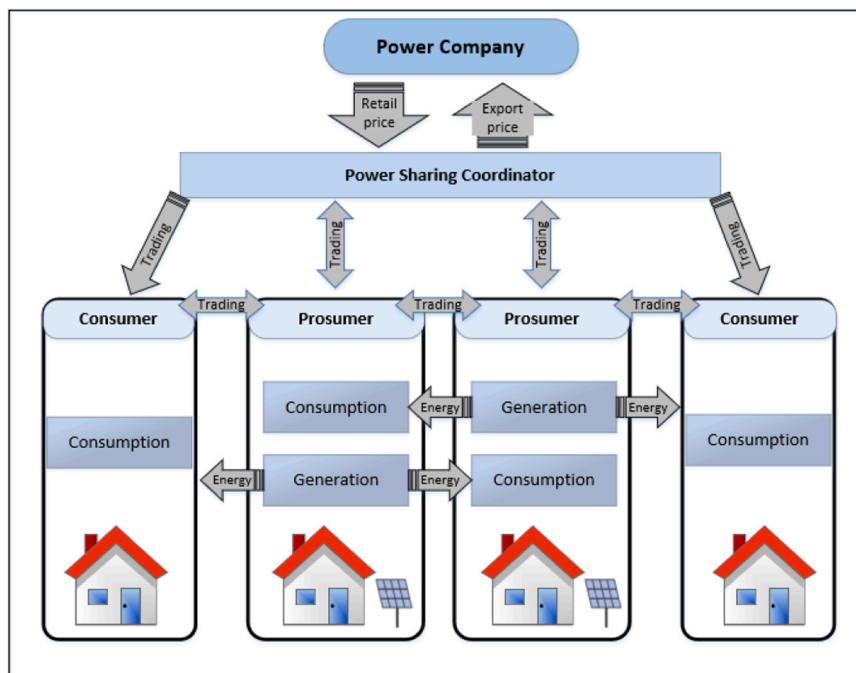


Fig. 1. Peer to Peer Energy Trading.

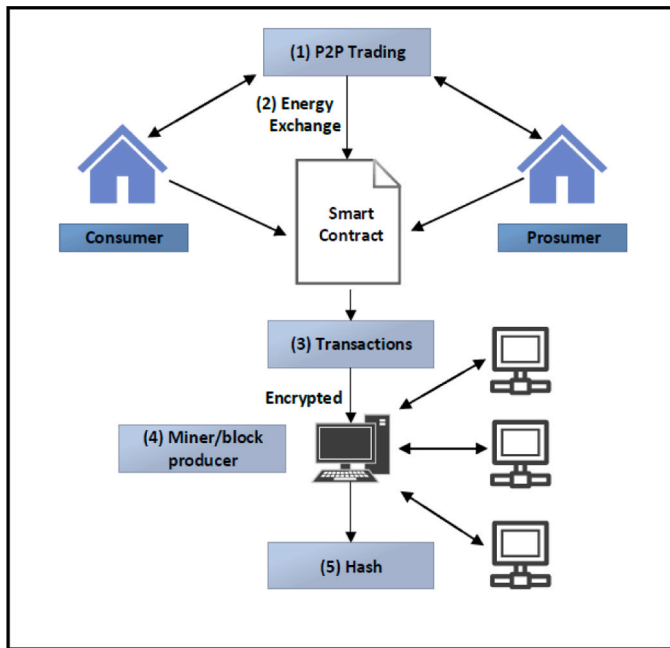


Fig. 2. Blockchain based Energy Transaction.

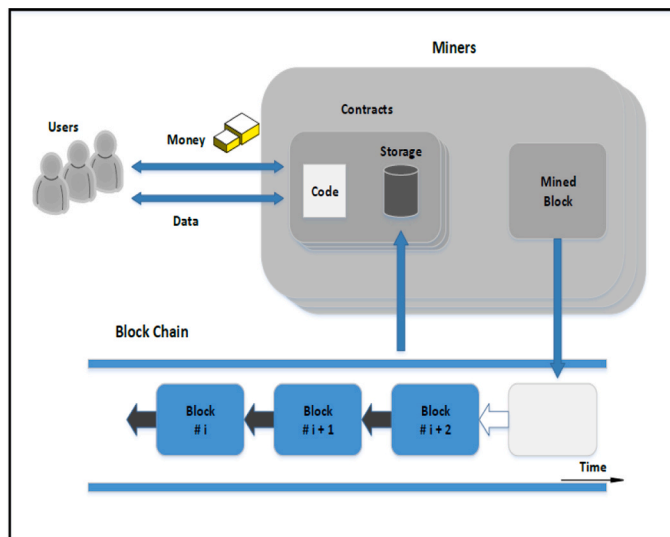


Fig. 3. Smart Contract System.

contract system. Every contract will have a 20-byte unique address assigned to it. Once the contract is deployed on the blockchain, the code of the contract is immutable and cannot be altered. In order to execute a contract, users can conveniently transmit a transaction to the designated address of the contract. Subsequently, all consensus nodes, also known as miners, in the network will execute this transaction in order to get a unanimous agreement on its outcome. The contract's state will thereafter be modified correspondingly. The contract has the ability to read from and write to its private storage, deposit money into its account balance, send and receive messages or money from users or other contracts, and also has the capability to establish new contracts (Delmolino et al., 2016; Zheng et al., 2020).

3.3. Web3 protocol

The emergence of Web3 had a significant impact on the cryptocurrency markets and blockchain communities since 2020. Web3 is set of libraries that enable to connect with Ethereum node remotely or locally. Essentially, it offers an Application Programming Interface (API) that allows to conveniently interact with the blockchain. The main concept is to offer distributed internet services without involving trusted third parties (TTP), which allows consumers to have greater autonomy (Eric Glen Weyl and Buterin, 2022). The fundamental principle of Web3 apps is to have complete control over their data, including, tokens, ownership, and other aspects, instead of relying on centralized organizations for managing.

The Web3 protocol primarily encompasses three distinct roles: the Web3 user, blockchain maintainer and service provider. The Web3 user is an individual who possesses ownership of data and can initiate a request by transmitting a transaction from their client. The service provider provides the user with an interface that operates directly on the blockchain and is capable of processing requests and interacting with blockchain platforms and the blockchain maintainer facilitates an operational framework for smart contracts that handle the processing of business operations and data storage (Wang et al., 2022).

4. System architecture

We have proposed a novel decentralized energy trading system established using smart contracts deployed on Ethereum network. The system utilizes ERC20 tokens to represent energy credits, facilitating trading between prosumer and consumer within a microgrid. By integrating the Web3 framework, the blockchain and microgrid are connected for efficient energy trading managed by smart contracts. The system simulates different times of day to dynamically manage the demand and supply, optimizing the microgrid's efficiency and adaptability. Energy trading is executed automatically according to predefined rules written smart contract.

4.1. Microgrid design

A simple microgrid system is designed in Simulink for distributed energy trading as shown in Fig. 4. The fundamental components of microgrid are load, solar array system and Energy management system (EMS). Specifications of microgrid used in simulation are given in Table 1. The load component quantifies the amount of energy utilized inside a microgrid, encompassing the electrical demand from residences, industries, and other linked entities. Simultaneously, the creation of solar arrays amplifies the integration of renewable energy into the system facilitated by panels, hence reducing reliance on power plants and the EMS manages the energy management in microgrid and also calculates cost of electricity at every instance. The data extracted for microgrid for specific time of day is connected to Ethereum by web3 library in python. Implementing Blockchain technology in microgrid architecture enhances security and transparency. The microgrid records all transactions and state changes on a blockchain platform guaranteeing a permanent and unalterable record of events. This approach to energy management and trading promotes effectiveness and long-term viability.

4.2. Energy token contract

The ERC-20 standard has been recognized as the predominant Ethereum standard for convertible tokens. Tokens that implement the ERC-20 interface have the ability to work together with numerous internet-based applications and smart contracts on the Ethereum platform. ERC-20 security vulnerabilities have garnered significant attention in recent years, mostly due to their extensive use and escalating

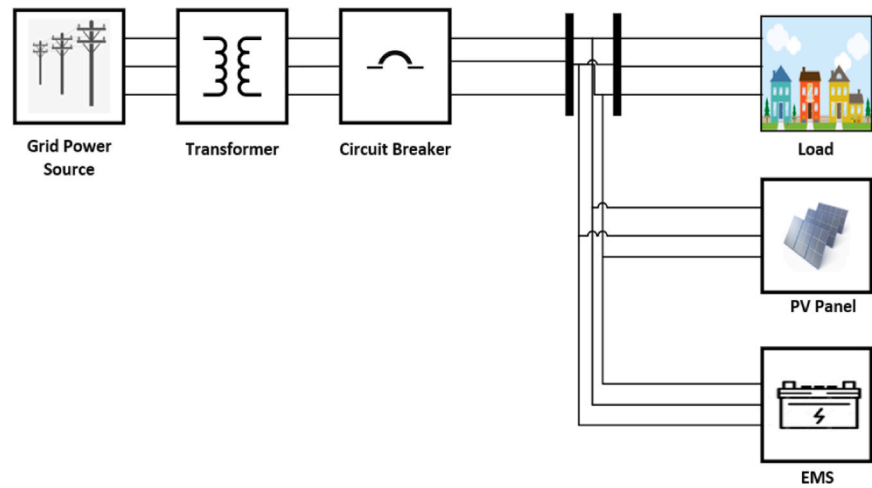


Fig. 4. Microgrid.

Table 1
Specifications of microgrid.

3 Phase Source	13.8 kV
Transformer	Primary winding: 13.8 kV Secondary winding:5 kV
Load	550 kW
Solar Array	500 kW
Battery	2500kWh

value (Ivanov et al., 2021). Presence of these tokens plays an important role in this project for decentralized energy trading system. This platform ensures transparent system for exchange and record of energy ownership to sell and purchase credits. This contract serves as fundamental framework for energy exchange. The pseudocode of algorithm is described below

Algorithm 1. .

```
1. State Variable
2. Events
   event Transfer
   event Approval
3. Constructor
4. Token Minting
   Ensure that target account is not the zero address
   Increase the total supply by minted amount
   Increase the balance of account by minted amount
5. Token Burning
   Ensure that target account is not the zero address
   Ensure account has enough token to burn
   Decrease the balance of account by the burned amount
   Decrease the total supply by the burned amount
6. Token Transfer
   If balances[sender] >= amount and to is not zero address then
     Deduct the tokens from sender's balance
     Add the tokens to recipient's balance
     Emit the transfer event
     Return True to indicate success
   Else
     Return False to indicate failure
7. Token Allowance
   Set the allowance [owner][spender] to amount
   Emit approval event from owner to spender with amount
8. Token Transfer From one account to another
   If balances[from] >= amount
     allowances[from][sender] >= amount
     from and to is not zero address
   Then
     Decrease balances[from] by amount
     Increase balances[to] by amount
     Decrease allowances[from][sender] by amount
     Emit transfer event: 'from' to 'to' with amount
     Return True
   Else
     Return False
9. Total Token Supply
```

Token supply is initiated by constructor function by providing initial supply of tokens then tokens are created and assigned to accounts by mint function, which means new tokens can be created (minted) and distributed to participants. For instance, households might receive tokens as an incentive for generating solar energy. Specific tokens are destroyed by burn function to reduce the total supply, maintaining a balanced token economy and then transfer from sender to recipient account by transfer function. Participants can transfer tokens between each other as payment for energy. Approve function in contract allows withdrawal of tokens for spender from owner's account and if allowed by owner then tokens are transferred by transfer function to recipient account and finally total supply function recovers the total supply of tokens in circulation.

4.3. Demand response contract

Demand response has emerged as a pivotal approach in field of energy management. These strategies urge consumers to reduce their electricity consumption during peak demand hours in exchange for incentives or prizes. The objective is to reduce grid stress, decrease dependency on non-renewable energy sources, and improve energy efficiency (Irtija et al., 2020). By means of blockchain technology and smart contracts, this project intends to create a platform for transparent energy trading in a decentralized network. This project offers not just efficient energy consumption, but also a balanced and secure environment for participants, that leads to the formation of a more resilient and effective environment. The proposed contract is written in solidity language on Ethereum platform. Following is pseudocode of proposed algorithm.

Algorithm 2.

“IERC20” interface defines standard events and functions and ERC20 contracts implement these functions to create complaint tokens. To manage the energy trading and demand response operations within the microgrid demand response contract is written in Solidity language. “State variable” function is to store essential data such as energy prices, energy credits, token balances, and contract ownership. To log important actions and changes within the contract, “Event log” function is introduced. These logs are stored on the blockchain and can be accessed later for auditing, debugging, or triggering other actions. The purpose of “Constructor” is to initialize the contract, setting the initial energy price and the contract deployer as the owner. “Modifier” is introduced to restrict access to certain functions to the contract owner, ensuring that only authorized users can perform sensitive actions. In this contract microgrid parameters are established by function “set microgrid parameters” so that they can only be accessible by owner and it also update the state variables. “Calculate reward” function calculate reward of updated parameters and emits the demand response event. Here reward refers to quantitative measure of system's efficiency and performance and this event comprises the information related to energy load, time-stamp, efficiency of EMS, cost of electricity, solar array generation and the calculated reward.

$$\text{Energy Load} = \text{Energy Credits} - \text{Energy Consumption}$$

$$\text{Solar Efficiency} = \frac{\text{Energy Generated}}{\text{Solar Array Capacity}}$$

$$\text{EMS Efficiency} = \frac{\text{Energy Saved}}{\text{EMS Capacity}}$$

$$\text{Load Factor} = \frac{\text{Energy Load}}{\text{Load Threshold}}$$

-
1. **IERC20 Interface**
ERC-20 Token Contract
Implement IERC20 functions
 2. **Demand Response Contract**
 3. **State variables**
 4. **Create Events**
 5. **Constructor**
Initiate Energy Price
Set the contract deployer as the owner
 6. **Modifier**
Access restricted to only owner
 7. **Set Microgrid Parameters**
 8. **Calculate Reward**
 9. **Trade Energy**
Verify seller has sufficient energy credits
Calculate total cost of trade
Ensure buyer has enough token and allowance
Deduct the traded amount from seller's energy credit
Add the traded amount to buyer's energy credit
Transfer the tokens from buyer to seller as payment
 10. **Set Energy Price**
New price
Ensure price is valid
Update energy price
 11. **Purchase Energy**
Transfer the tokens from the buyer to the contract as payment
Add the purchased amount to the buyer's energy credits
Emit an event to log the energy purchase
 12. **Calculate Reward**
 13. **Calculate Efficiency**
 14. **Calculate load and cost**
 15. **Capping factors**
 16. **Calculate total reward**
Total Reward = base reward + solar efficiency + ems efficiency – load factor – cost factor
-

$$\text{Cost Factor} = \frac{\text{Energy Price} * \text{Energy Consumption}}{\text{Cost Threshold}}$$

Then trading of energy is emitted by “trade energy” function which ensures that seller has enough energy credits and buyer has enough tokens and allowance for transactions. “Set energy price” function in smart contract, set the price of energy for microgrid and is only accessible by contract’s owner then “purchase energy” function verifies that buyer has sufficient token and transfer them from buyer to contract and increases seller energy credits. After that function “calculate reward” again states a base reward value and then efficiency of energy generated by solar array and EMS are calculated. Cost and load factors are also computed. Then calculated factors are capped which means those fac-

4.4. Methodology

Ganache is a local blockchain designed for Ethereum development. It provides private environment for developers to test their contracts. In this project, we have used web3 library in python to establish connection with Ethereum node hosted by Ganache. After successful connection, the proposed script proceeds to define addresses and ABI (application binary interfaces) of Energy token and Demand Response contracts. Web3 then instantiated these contracts and enable the algorithm to interact with their functions. Pseudocode of proposed algorithm is mentioned below

Algorithm 3.

```

1. Web3 library to connect with Ethereum
   w3 = Web3 (HTTP Provider (ethereum node url))
2. Contract addresses and ABI
3. Extract microgrid data
4. Set up accounts
5. Create contract instances
6. Interact with smart contract
7. Energy Token class
   Method __init__ (self, token _ contract, owner _ account, demand _ response _ contract)
   Method get _ balance
       Convert address to checksum format
       Call and return token contract's balance of method with the address
   Method transfer _ tokens
       Convert to _ address to checksum format
       Get nonce of self. account
       Build transaction for transferring tokens
       Sign the transaction with self. account's private key
       Send signed transaction
       Wait for and return transaction receipt
   Method trade _ energy
       Get current energy price
       Calculate total cost as amount * price per unit
       If price per unit > current energy price:
           Set new energy price to price per unit
           Determine transfer percentage
       If price per unit < 90;
           Token Transfer = Surplus Energy* 10%
       If price per unit ≥ 90;
           Token Transfer = Surplus Energy* 5%
       Calculate amount to trade
       Calculate total cost
       Approve self-owner account for total cost
       Transfer tokens from buyer to seller
8. Simulate time period
   Calculate available solar energy and energy demand
   If excess energy > 0
       Calculate amount to trade (minimum of excess energy and amount)
       Calculate required amount (amount to trade price per unit)
       Get buyer's balance
       If buyer balance < required amount
           Print insufficient balance message
       Else
           Transfer tokens from buyer to seller
           Print buyer's balance after trade
   Else
       No excess energy
9. Instantiate the Energy Token class with demand response contract
10. Check buyer's balance
11. Perform time period simulation
    "Morning, Noon, Evening"

```

tors which cross the threshold value are limited to prevent excessive impact on the reward. In the end total reward is calculated. Proposed demand response system is collectively formed from these factors which allows dynamic adjustments for parameters of microgrid, trading, adjusting energy prices and rewarding the participants. This smart contract ensures that all operations are restricted to owner only.

In proposed algorithm Web3 interface is used to create connection between blockchain and microgrid data. First instances of both contracts are created based on their addresses and ABIs. Then energy token class is designed to perform to execute activities such as balance checking, token transfer, and energy trading simulation over various time periods. the "trade energy" method works according to following formulas

status	0x1 Transaction mined and execution succeed
transaction hash	0x01741525061bb51e852a94bd70226dc80b8fc0b999c8de803bc239035a0d3ee4 
block hash	0x2f68f37182003391c052fbc3f0ff59234767ad134815ddbcf9ed48a8b8b392b0 
block number	102 
contract address	0x62f1464e643fb53fe6058cca4f2401c81929443d 
from	0xafdddebb6c904bce00946b320cc6bf76bdbd28e87 
to	EnergyToken.(constructor) 
gas	1026179 gas 
transaction cost	1026179 gas 
input	0x608...40000 

Fig. 5. Energy token contract.

$$\text{Amount to Trade} = \text{Surplus Energy} * \text{Transfer percentage}$$

$$\text{Total Cost} = \text{Amount to Trade} * \text{New Energy Price}$$

Where,

Surplus energy: Energy beyond demand

Transfer percentage: Percentage of surplus energy that will be traded

Amount of energy: The quantity of energy being traded

New energy price: updated price per unit of energy for the trade

Different time periods of day are simulated by function “simulate_time_periods”, it considers microgrid data and transfers tokens between accounts based on excess energy availability and calculates the overall microgrid characteristics. The procedure begins with the utilization of the accessible solar energy and the corresponding energy requirements. If the amount of generated energy surpasses the energy needed, it indicates that there is surplus energy. This method then determines the quantity of energy to be traded. The buyer's balance is verified during this step to ensure that it reaches or surpasses the required amount. If the buyer's balance is insufficient, a notice is printed stating "insufficient balance". If the amount is adequate, tokens are transferred from the buyer to the seller.

5. Results and discussion

In this section, we present the outcomes of our simulation of the energy trading dynamics within a microgrid that utilized blockchain technology. The simulation was run on the Ethereum blockchain and demonstrated the interaction between energy producers, customers, and a demand response system implementing Solidity smart contracts. Practicality and effectiveness of energy trading based on blockchain were the primary objectives.

The data collection for our blockchain-based energy trading system is derived from a microgrid simulation model developed in Simulink. This data contains multiple metrics that illustrate the response of the microgrid at different times of day - morning, noon, and night time. In order to comprehend the dynamic nature of microgrid operation at any given instant, the dataset comprises of every record with specific time units measured during simulation. In the energy trading logic, the energy management system and solar array play critical roles. When the energy provided by the EMS and solar array combined exceeds than the demand of the microgrid, excess power can be traded, establishing a self-sustaining system for the exchange of electrical energy.

status	0x1 Transaction mined and execution succeed
transaction hash	0xbb20865688379ea44274782a2a22b12236d70bf6f143a37bffe5232a26e11993 
block hash	0x51df8b3a918d459d4a8f7237a4255d3faa2a2c3dcfa87fd78775f5257e3065c5 
block number	3 
contract address	0xbe857f81ac123053c4413c977fc4daa898fc0889 
from	0xafdddebb6c904bce00946b320cc6bf76bdbd28e87 
to	DemandResponse.(constructor) 
gas	1403329 gas 
transaction cost	1403329 gas 
input	0x608...9ea83 

Fig. 6. Execution of Demand Response Contract.


```
Microgrid Data: Time: 21600.0, Load: 30000, Solar Array: 10000, EMS: 20000, Cost: 50
No excess energy available for trade.
Microgrid Data: Time: 21615.0, Load: 32000, Solar Array: 15000, EMS: 21000, Cost: 55
```

(a) Simulation of Morning time

```
Simulating noon...

Microgrid Data: Time: 43215.0, Load: 61000, Solar Array: 550000, EMS: 31000, Cost: 85
Transfer successful. 2500 tokens transferred from 0xafDdeb86c904BcE0946b320C6bF76bdb28e87 to 0x9F7cdC422981ABe3D716949dEf5A40B5618CaC0.
Transaction Receipt: AttributeDict({'transactionHash': HexBytes('0x423ecb12719d85c96576074872b38df4d58d54799d43b80c32b018b688ce6e58')})
Trade successful. 2500 tokens transferred from 0xafDdeb86c904BcE0946b320C6bF76bdb28e87 to 0x9F7cdC422981ABe3D716949dEf5A40B5618CaC0.
Microgrid Data: Time: 43245.0, Load: 63000, Solar Array: 65000, EMS: 33000, Cost: 95
Transfer successful. 1750 tokens transferred from 0xafDdeb86c904BcE0946b320C6bF76bdb28e87 to 0x9F7cdC422981ABe3D716949dEf5A40B5618CaC0.
Transaction Receipt: AttributeDict({'transactionHash': HexBytes('0x2b67f0be42d36c7ff8549725b4c2a48ec7776147d36bc3b250e63b068dbb321')})
Trade successful. 1750 tokens transferred from 0x77692bAd054Ae81075df1cd67c71e89C9e5f8f to 0x9F7cdC422981ABe3D716949dEf5A40B5618CaC0.
```

(b) Simulation of Noon time

```
Simulating evening...
Microgrid Data: Time: 64800.0, Load: 50000, Solar Array: 40000, EMS: 30000, Cost: 85
No excess energy available for trade.
Microgrid Data: Time: 64815.0, Load: 51000, Solar Array: 35000, EMS: 29000, Cost: 90
No excess energy available for trade.
```

(c) Simulation of Evening time

Fig. 9. (a) Simulation of Morning time. (b) Simulation of Noon time. (c) Simulation of Evening time.

modify price of energy, if needed. This permits for changes provided by this feature provides flexibility and reactivity to changing situations. The function “Energy Purchase” in contract allows users to participate in microgrid ecosystem by receiving corresponding energy credits. The calculate Reward function is an intriguing feature that automatically calculates rewards based on different efficiency and load or cost considerations. In nutshell, this contract provides financial prudence, security and transparency to stakeholders.

The successful deployment of this contract is shown in Fig. 6. Smart contract automatically executed in two conditions.

5.2.1. During energy trading

When excess energy is available for trading the algorithm calculates the amount of energy to trade and total cost. It then interacts with smart contract to transfer tokens from buyer to seller. This process involves building, signing and sending the transaction to Ethereum network. The transaction is executed automatically once it is sent to the network and smart contract handles the transfer of tokens.

5.2.2. During energy purchase

When the buyer initiates energy purchase, energy token transfer is executed automatically by “purchase energy” function written in smart contract.

5.3. Microgrid simulation

This model reflects various facets of the microgrid’s functionality, including microgrid’s load energy, solar array output power and EMS contributions as well as price of electricity. Fig. 7 shows the power (kW) graph of photovoltaic generation, Energy storage system, grid and load. The graph demonstrates that

- The yellow graph depicts the solar generation derived from photovoltaic (PV) panels. The value begins at zero and steadily rises, reaching its highest point at the middle, before gradually declining again. The peak point indicates that the highest amount of solar power is generated at midday, when sunlight is at its peak.
- The blue graph indicates the energy storage system. It starts from below zero and then steps up to around 200 kW before stepping down again. This pattern indicates charging and discharging of energy storage systems.
- The grid power is represented by green graph, it fluctuates between –500 kW and 1000 kW. These functions indicate shifts in the demand for and supply of electricity from the grid.
- The purple line represents the evolution of power usage across time. The changing shape of the graph indicates that power consumption varies during the day. The load curve’s peaks indicate the times of day when power demand is at its peak, such as early morning and late at night. The PV generation curve is out of phase with the PV generation curve. During the peak hours of solar generation, the load is quite low. On the other hand, the load is typically higher in the morning and evening, when PV generation is lower.

P2P energy trading has been effectively executed by the smart contracts. Energy trading is only successful when excess energy is available from solar array and EMS and it surpasses the current demand within the microgrid. Fig. 8 shows that buyer has sufficient balance required for trading. The simulation results of specific time periods are displayed for detailed analysis of the microgrid’s performance within a day. Due to diminished sunshine exposure the solar generation is typically reduced during morning, resulting in insufficient output from the solar array to meet the energy demand of the microgrid during this specific time period¹. The morning energy consumption may be significantly elevated

due to additional activities such as waking up, preparing for the day, and using household appliances. Simulation indicates that no excess energy is available for trading during morning period Morning because EMS and solar energy did not surpass the energy demand as shown in Fig. 9 (a). The comprehensive simulation spans a full 24-hour period, Figures only illustrates a representative subset of the entire dataset. However, from Fig. 9(b), it can be seen that successful trading happened at Noon as combined solar energy and EMS generation exceeded the energy demand, leading to a surplus available for trade. Trading does not occur when there is surplus of energy accessible, but the buyer lacks sufficient balance at that moment. Similar to morning period, during evening, no excess energy is available for trading due to reduced solar energy generation by the solar array and lower contribution from the EMS as illustrated in Fig. 9(c).

To further analyze the simulation results, calculations of three different scenarios are presented.

Time: 21,600.0 (Morning)

Load: 30,000 units

Solar Array: 10,000 units

EMS: 20,000 units

Cost per Unit: 50

Total available energy = Solar Array + EMS = 10,000 + 20,000 = 30,000 units

Energy Surplus = Total available energy – Load = 30,000–30,000 = 0

For Time: 43215.0 (Noon)

Load: 61,000 units

Solar Array: 55,000 units

EMS: 31,000 units

Cost per Unit: 85

Total available energy = Solar Array + EMS = 55,000+31,000=86,000 units

Energy Surplus = Total available energy – Load = 86,000–61,000=25,000 units

Transfer Percentage: Since the cost per unit (85) is less than 90, use 10 %.

Token Transfer = Energy Surplus * Transfer Percentage = 25,000 * 10 % = 2500 tokens

Time: 64,845.0 (Evening)

Load: 53,000 units

Solar Array: 25,000 units

EMS: 27,000 units

Cost per Unit: 100

Total available energy = Solar Array + EMS = 25,000+27,000=52,000 units

Energy Surplus = Total available energy – Load = 52,000–53,000= –1000 units

6. Conclusion

A comprehensive novel approach is presented in this paper to revolutionized energy trading within microgrids through integration of blockchain technology and smart contracts. Energy token and demand response contracts in decentralized peer to peer energy trading enhance security, efficiency and transparency in microgrid operation. The introduction of digital tokens facilitates transparent tracking and equitable profit distribution. By eliminating central authorities in energy systems, this approach improves reliability of financial records and reduces the potential of system failure and also results in cost reduction and empowerment of consumers to have control on their power resources. Moreover, P2P network facilitates load balancing and ensure efficient energy sharing amongst participants. Smart contracts, written in solidity language are deployed on Ethereum platform, when specific conditions mentioned in contract are met, they execute automatically.

The simulations results showed microgrid's performance during specific time periods of a day. Significantly, energy trading is not

conducted during the morning and evening hours due to insufficient solar power generated by the array along with reduced input from the Energy Management System (EMS). However, at noon when excess energy exceed demand, successful trading is executed. The demand-response mechanism ensures that transactions occur during periods of surplus energy, demonstrating the efficiency and reliability of the proposed peer-to-peer energy trading system. From calculations, it can be seen that no extra energy is available for trading at time 21,600 s, while 2500 tokens are traded between buyer and seller at time 43,215 s and there is a deficit of 1000 units in energy supply at time 64,845 s, indicating that the amount of energy available is insufficient to meet the demand. As a result, there is no surplus energy available for trading at that time.

The proposed research on blockchain based microgrid system presents a promising future scope and few limitations. Ensuring compliance with energy regulations is crucial for successful implementation of this framework. Moreover, limitations related to energy storage capacity and availability could impact overall performance of the system. Future exploration can focus on blockchain interoperability and integration with multi-microgrid system. Additionally, adaptive pricing models that respond to real time supply and demand dynamics could improve system responsiveness and efficiency. In nutshell, this research not only presents a theoretical framework but also demonstrates the practical implementation of a novel paradigm for demand-responsive for energy trading within microgrids through comprehensive simulations.

Ethical considerations

This research was conducted in accordance with ethical standards.

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CRediT authorship contribution statement

Nimrah Saeed: Writing – original draft, Methodology, Conceptualization. All authors have contributed significantly to the research and manuscript preparation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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